① X is Cauchy r.v with parameter α

to show $\Phi_X(\omega) = e^{-\alpha|\omega|}$, where $\Phi_X(\omega) = E[e^{j\omega x}]$

is the characteristic function of X.

pdf or prob density f^n of X, centered at 0

$$f_X(x) = \frac{\alpha}{\pi(x^2+\alpha^2)}, \quad -\infty < x < \infty \quad \& \quad \alpha > 0 \quad \text{—— ①}$$

The characteristic function of X given

$$\Phi_X(\omega) = E\left[e^{j\omega X}\right]$$

$$= \int_{-\infty}^{\infty} e^{j\omega x} f_X(x)\,dx$$

if X has pdf of $f_X(x)$ then for any f^n $g(X)$

$E[g(X)] = \int_{-\infty}^{\infty} g(x) f_X(x)\,dx$

here $g(X) = e^{j\omega x}$

$$= \int_{-\infty}^{\infty} e^{j\omega x} \frac{\alpha}{\pi(\alpha^2+x^2)}\,dx = \frac{\alpha}{\pi}\int_{-\infty}^{\infty} \frac{e^{j\omega x}}{x^2+\alpha^2}\,dx \quad \text{—— ②}$$

$e^{j\omega x} = \cos(\omega x) + j\sin(\omega x)$

$\cos(\omega x)$ is even f^n

$\sin(\omega x)$ is odd f^n

Subst^n in ② we get

$$= \frac{\alpha}{\pi}\left[\int_{-\infty}^{\infty} \frac{\cos\omega x}{x^2+\alpha^2}\,dx + \int_{-\infty}^{\infty} \frac{\sin\omega x}{x^2+\alpha^2}\,dx\right]$$

$\frac{1}{x^2+\alpha^2}$ is even

$$= \frac{\alpha}{\pi}\left[\int_{-\infty}^{\infty} \frac{\cos\omega x}{x^2+\alpha^2}\,dx\right]$$

since $\int_{-\infty}^{\infty} \frac{\sin\omega x}{x^2+\alpha^2}\,dx = 0$ odd f^n.

$$= \frac{\alpha}{\pi}\int_{0}^{\infty} 2\int \frac{\cos\omega x}{x^2+\alpha^2}\,dx = \frac{2\alpha}{\pi}\cdot I$$

where $I = \int_0^{\infty} \frac{\cos\omega x}{x^2+\alpha^2}\,dx$

$$I = \int_0^\infty \frac{\cos \omega x}{x^2 + \alpha^2}\, dx = \frac{\pi}{2\alpha} e^{-a|\omega|}$$

$$\frac{dI}{d\omega} = \int_0^\infty$$

$$\therefore \quad \Phi_X(\omega) \quad \frac{2\alpha}{\pi} \cdot \frac{\pi}{2\alpha} e^{-\alpha|\omega|} = e^{-\alpha|\omega|} \quad \text{proved}$$

Another way, to solve integration

$$\Phi_X(\omega) = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x)\, dx$$

$$= \frac{\alpha}{\pi} \int_{-\infty}^{\infty} e^{j\omega x} \cdot \frac{1}{x^2 + \alpha^2}\, dx$$

we consider $g(\omega) = e^{-\alpha|\omega|}$ then FT of $g(\omega)$ is

$$G(x) = \int_{-\infty}^{\infty} e^{-j\omega x} g(\omega)\, d\omega$$

$$= \int_{-\infty}^{\infty} e^{-j\omega x} e^{-\alpha|\omega|}\, d\omega$$

$$= \int_{-\infty}^{0} e^{-j\omega x} e^{+\alpha\omega}\, d\omega + \int_0^{\infty} e^{-j\omega x} e^{-\alpha\omega}\, d\omega$$

$-\infty < \omega < 0 \quad |\omega| = -\omega$

$0 < \omega < \infty \quad |\omega| + \omega$

$$= \int_{-\infty}^{0} e^{(\alpha - jx)\omega}\, d\omega + \int_0^{\infty} e^{-(\alpha + jx)\omega}\, d\omega$$

$$= \left. \frac{e^{(\alpha - jx)\omega}}{\alpha - jx} \right|_{-\infty}^{0} - \left. \frac{e^{-(\alpha + jx)\omega}}{\alpha + jx} \right|_0^{\infty}$$

$$= \frac{1}{\alpha - jx} + \frac{1}{\alpha + jx} = \frac{2\alpha}{\alpha^2 + x^2}$$

now inverse FT $g(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} G(x)\, e^{j\omega x}\, dx$

$$\therefore g(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{2\alpha \cdot e^{j\omega x}}{\alpha^2 + x^2}\, dx$$

$$e^{-\alpha|\omega|} = \frac{1}{\pi}\int_{-\infty}^{\infty} \frac{\alpha\, e^{j\omega x}}{\alpha^2 + x^2}\, dx$$

$$e^{-\alpha|\omega|} = \int_{-\infty}^{\infty} \frac{\alpha \cdot e^{j\omega x}}{\pi(\alpha^2 + x^2)}\, dx = \int_{-\infty}^{\infty} e^{j\omega x} f_X(x)\, dx$$

where $= E\left[e^{j\omega x}\right]$

$e^{-\alpha|\omega|} = \Phi_X(\omega)$ hence proved.

(2) X is r.v with characteristic f^n $\Phi_X(\omega)$. To show $|\phi_X(\omega)|$ takes on its max at $\omega = 0$

Characteristic f^n defined as $\phi_X(\omega) = E\left[e^{j\omega X}\right]$

$$= \int_{-\infty} e^{i\omega x} f_X(x)\, dx$$

where $f_X(x)$ is prob density f^n of X

$$|\phi_X(\omega)| = \left|\int_{-\infty}^{\infty} e^{i\omega x} f_X(x)\, dx\right| \quad \text{—— (1)}$$

We know,

$(|a+b| \leq |a| + |b|)$ $\quad \therefore \left|\int (\cdot)\, dx\right| \leq \int |\cdot|\, dx$

$\therefore$ (1) becomes

$$|\phi_X(\omega)| = \left|\int_{-\infty}^{\infty} e^{i\omega x} f_X(x)\, dx\right| \leq \int_{-\infty}^{\infty} \left|e^{i\omega x} f_X(x)\right| dx$$

$$|e^{i\omega x}| = \left\{\cos^2(\omega x) + \sin^2(\omega x)\right\}^{1/2} = \sqrt{1} = 1$$

$$\therefore \left| \int_{-\infty}^{\infty} e^{i\omega x} f_X(x)\,dx \right| \leq \int_{-\infty}^{\infty} \left| e^{i\omega x} f_X(x) \right| dx$$

$$= \int_{-\infty}^{\infty} \left| e^{i\omega x} \right| \left| f_X(x) \right| dx$$

$$= \int_{-\infty}^{\infty} f_X(x)\,dx$$

$f_X(x) \geq 0$ & $|e^{i\omega x}| = 1$

$$\therefore \int_{-\infty}^{\infty} f_X(x)\,dx = 1.$$

$$\therefore |\phi_X(\omega)| \leq 1.$$

when $\omega = 0$, $\phi_X(\omega = 0) = E\left[e^{j(0)X}\right] = E[e^0] = 1.$

$$\therefore |\phi_X(0)| = 1$$

$\therefore$ Since $|\phi_X(\omega)| \leq 1 \quad \forall \omega \in \mathbb{R}$

& $|\phi_X(0)| = 1.$

$\therefore$ max mag of char f^n occurs at $\omega = 0$

Ans.

(3) Use moment theorem to find mean and variance of poisson r.v. X.

MT states that $E[X^n] = \phi_X^{(n)}(0) = \left.\dfrac{d^n \phi_X(s)}{ds^n}\right|_{s=0}$

where $\phi_X(s) = E\left[e^{sX}\right]$, $s \in \mathbb{R}$

is the moment generating function for a poisson r.v X

we know $P(X=k) = \frac{\lambda^k e^{-\lambda}}{k!}$ is pmf of poisson

$$\therefore \quad \phi_X(s) = E[e^{sX}] = \sum_{k=0}^{\infty} e^{sk} P(X=k)$$

$$= \sum_{k=0}^{\infty} e^{sk} \frac{\lambda^k e^{-\lambda}}{k!}$$

$$= e^{-\lambda} \sum_{k=0}^{\infty} \frac{(\lambda e^s)^k}{k!}$$

let $u = \lambda e^s$

$$= e^{-\lambda} \left(\sum_{k=0}^{\infty} \frac{u^k}{k!} \right)$$

$\therefore$

$$= e^{-\lambda} e^{u}$$

$$= e^{-\lambda} e^{\lambda e^s}$$

$$\phi_X(s) = e^{\lambda(e^s - 1)}$$

$$E[X] = \left. \frac{d\phi_X(s)}{ds} \right|_{s=0}$$ from moment theorem.

$$= e^{\lambda(e^s - 1)} \cdot (\lambda e^s) \Big|_{s=0}$$

$$= e^{\lambda(e^0 - 1)} \cdot (\lambda e^0)$$

$$= e^{\lambda \cdot 0} \cdot \lambda = \lambda$$

$\therefore \quad E[X] = \lambda$ this is mean

$$Var(X) = E[X^2] - (E[X])^2 = \left. \frac{d^2\phi_X(s)}{ds^2} \right|_{s=0} - \left. \frac{d\phi_X(s)}{ds} \right|_{s=0}$$

$$= \left. \frac{d^2\phi_X(s)}{ds^2} \right|_{s=0} - \lambda^2$$

$$\frac{d^2\phi_X(s)}{ds^2} = \frac{d}{ds}\left(\lambda e^s \cdot e^{\lambda(e^s-1)}\right)$$

$$= \lambda e^s \frac{d}{ds}\left[e^{\lambda(e^s-1)}\right] + e^{\lambda(e^s-1)} \cdot \frac{d}{ds}(\lambda e^s)$$

$$= \lambda e^s \, \lambda e^s e^{\lambda(e^s-1)} + e^{\lambda(e^s-1)} \cdot \lambda e^s$$

$$= \lambda e^s e^{\lambda(e^s-1)} (\lambda e^s + 1)$$

at $s = 0$

$$\left.\frac{d^2\phi_X(s)}{ds^2}\right|_{s=0} = \lambda \cdot 1 \cdot (\lambda + 1) = \lambda^2 + \lambda$$

$$\therefore \quad Var(X) = (\lambda^2 + \lambda) - (\lambda)^2 = \lambda$$

$\therefore$ $Var(X) = \lambda$ Variance

$E[X] = \lambda$ mean

Ans

4 when $X \sim$ Exponential(λ), $\lambda > 0$

pdf $f_X(x) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & x < 0 \end{cases}$

$\phi_X(s) = E[e^{sX}]$ for continuous r.v

$$= \int_{-\infty}^{\infty} e^{sx} f_X(x)\,dx = \int_0^{\infty} e^{sx} \lambda e^{-\lambda x}\,dx$$

$$= \lambda \int_0^{\infty} e^{-(\lambda - s)x}\,dx$$

$$= \lambda \left[-\frac{1}{(\lambda - s)} \cdot \frac{1}{e^{(\lambda - s)x}} \right]_0^\infty$$

$$= \lambda \left[0 + \frac{1}{\lambda - s} \right]$$

$$= \frac{\lambda}{\lambda - s}$$

$$\therefore \phi_X'(s) = \frac{d}{ds}\left(\frac{\lambda}{\lambda - s}\right) = \lambda(-1)(\lambda - s)^{-2}(-1)$$

$$= \frac{\lambda}{(\lambda - s)^2}$$

at $s = 0$, $\quad E[X] = \frac{\lambda}{\lambda^2} = \frac{1}{\lambda}$

$$\phi_X''(s) = \frac{d}{ds}\left(\frac{\lambda}{(\lambda - s)^2}\right) = \lambda(-2)(\lambda - s)^{-3} \cdot (-1) = \frac{2\lambda}{(\lambda - t)^3}$$

at $s = 0$, $\quad = \frac{2\lambda}{\lambda^3} = \frac{2}{\lambda^2}$

$$\therefore Var(X) = E[X^2] - (E[X])^2$$

$$= \frac{2}{\lambda^2} - \left(\frac{1}{\lambda}\right)^2 = \frac{2}{\lambda^2} - \frac{1}{\lambda^2} = \frac{1}{\lambda^2}$$

$$\therefore \left.\begin{array}{l} Var(X) = \frac{1}{\lambda^2} \\ \text{mean } E[X] = \frac{1}{\lambda} \end{array}\right\} \text{ Ans}$$

5 let $Y = e^{X}$ & X is Gaussian r.v.

to find Y.

$X \sim$ gaussian r.v with mean μ & variance σ^2

pdf $f_X(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$, $-\infty < x < \infty$

Since $Y = e^{x}$, $0 < Y < \infty$ lets define $Y = g(X) = e^{X}$

$\therefore f_Y(y) = 0$ for $y \leq 0$

we have $Y = e^{X} = g(X)$ $\quad X = g^{-1}(y)$
$= \ln Y$

$\therefore \frac{d}{dy}(g^{-1}(y)) = \frac{d}{dy}(\ln y)$
$= \frac{1}{y}$

$\therefore$ ~~$f_Y(y) = \frac{f_X(g^{-1}(y))}{\frac{d\,g(y)}{dy}} \cdot e^{x}$~~

~~$= f_X$~~

$\therefore$ formula is $f_Y(y) = \frac{f_X(g^{-1}(y))}{1/y}$

$= f_X \ln y \cdot \frac{1}{y}$

or $= \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left[-\frac{(\ln y - \mu)^2}{2\sigma^2}\right] \cdot \frac{1}{y}$.

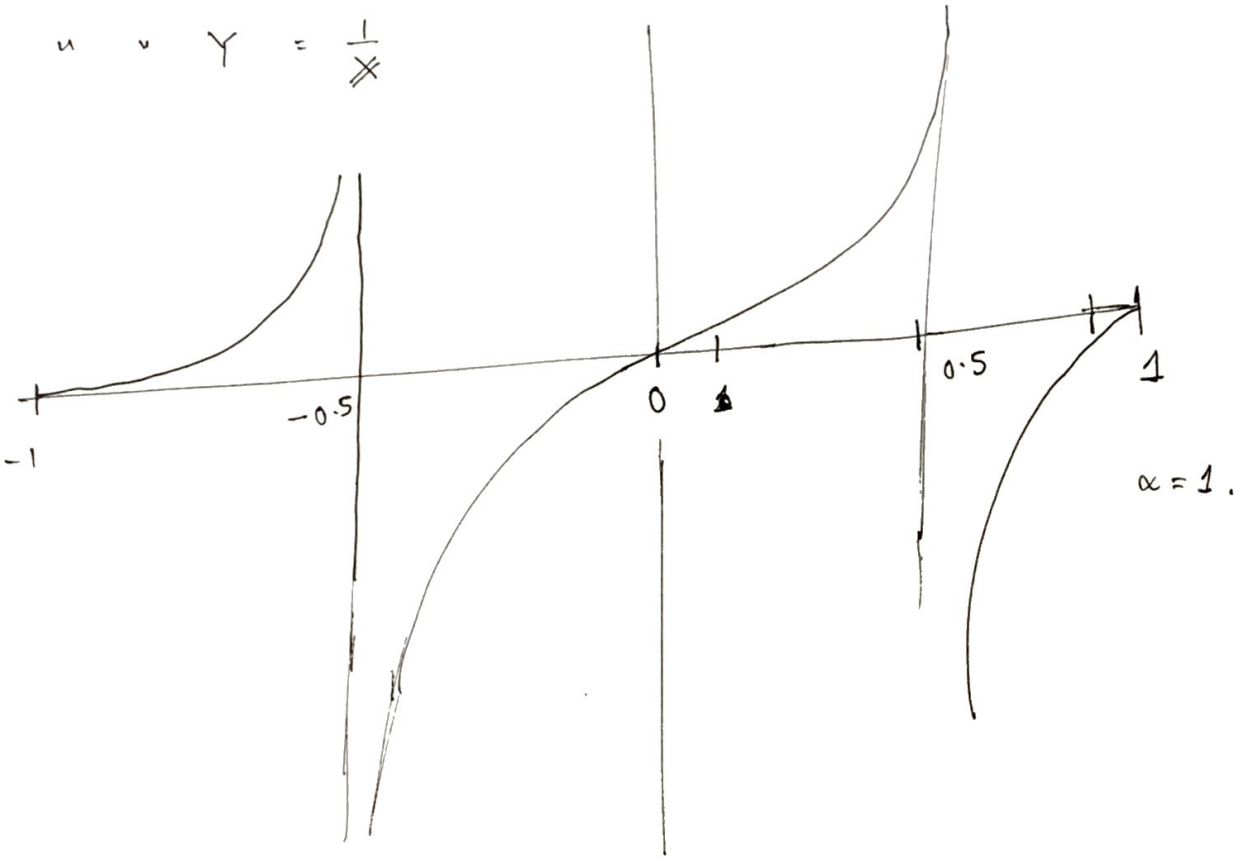
$\therefore f_Y(y) = \frac{1}{y\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln y - \mu)^2}{2\sigma^2}\right)$ Answer

6. X is uniformly distributed $[-1, 1]$

(a) find pdf of $Y = \alpha \tan \pi X$

(b) " " " $Y = \frac{1}{X}$

(a)



the pdf is $\begin{cases} \frac{1}{1-(-1)} = \frac{1}{2}, & -1 < x < 1 \\ 0, & \text{otherwise.} \end{cases}$

$Y = \alpha \tan(\pi X) = g(X) \quad \therefore \quad X = g^{-1}(Y)$

$y = \alpha \tan(\pi x) \qquad = \frac{Y}{\alpha} \tan(\pi X)$

$\frac{y}{\alpha} = \tan(\pi x) \quad \cancel{\pi X = \arctan\left(\frac{Y}{\alpha}\right) + k\pi}$

$\arctan\left(\frac{y}{\alpha}\right) = \pi x$

$\pi x = \arctan\left(\frac{y}{\alpha}\right) + k\pi \quad$ where $k \in \mathbb{Z}$

divide by π:

$x_k = g_k^{-1}(y) = \frac{1}{\pi} \arctan\left(\frac{y}{\alpha}\right) + k$

valid integers $k \in \mathbb{Z}$ are $k = -1, 0, 1$.

since $X \in [-1, 1]$ & $-1/2 < \frac{1}{\pi} \arctan(y/\alpha) < 1/2$

$k = -1, 0, 1$ for $x_{-1}(y)$, $x_0(y)$ & $x_1(y)$

$\therefore$ transformation

$$f_Y(y) = \sum_k f_X(x_k(y)) \left| \frac{d}{dy} x_k(y) \right|$$

$$x_k(y) = \frac{1}{\pi} \arctan(y/\alpha) + k$$

$$\frac{dx_k}{dy} = \frac{1}{\pi} \frac{1}{1+(y/\alpha)^2} \cdot \frac{1}{\alpha} \quad \text{or} \quad \frac{dx_k}{dy} = \frac{1}{\pi\alpha} \frac{1}{1+(y/\alpha)^2}$$

since $f_X(x_k(y)) = \frac{1}{2}$

$$f_Y(y) = \left(\frac{1}{2}\right) \cdot \frac{1}{\pi\alpha} \cdot \frac{1}{1+(y/\alpha)^2} \times \text{num of branches}$$

for any y there are two values of $x \in [-1, 1]$ that satisfy $y = \alpha \tan(\pi x)$. For example, if $y > 0$, the branches $k = 0$ [where $x \in (0, 0.5)$) and $k = 1$ where $x \in [0.5, 1)$ both contribute. if $y < 0$, the branches $k = -1$ where $x \in (-1, -0.5)$ and $k = 0$ where $x \in (-0.5, 0)$ contribute.

$$f_Y(y) = 2\left(\frac{1}{2} \cdot \frac{1}{\frac{\pi}{\alpha}(\alpha^2+y^2)}\right) = \frac{\alpha}{\pi(\alpha^2+y^2)}$$

$\therefore$ the pdf is $\dfrac{\alpha}{\pi(\alpha^2+y^2)}$, $-\infty < y < \infty$

(b). Let $Y = g(X) = \dfrac{1}{X}$ when $X \sim U(-1, 1)$

Since X is uniformly distributed between $[-1, 1]$, its pdf is

$$f_X(x) = \begin{cases} 1/2 & \text{if } -1 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

we define $Y = g(X) = \frac{1}{X}$ ~~we derivative~~

inverse transformation, $y = \frac{1}{x}$ or $x = \frac{1}{y}$

$$\therefore g^{-1}(y) = \frac{1}{y}$$

Since $X \in [-1, 1]$ if $0 < X \leq 1$ then $Y \geq 1$

if $-1 \leq X < 0$, then $Y \leq -1$

so $Y \in (-\infty, -1] \cup [1, \infty)$

$x = g^{-1}(y) = \frac{1}{y}$ $\qquad \frac{dy}{dx} = -\frac{1}{x^2}$

$g^{-1}(y) = \frac{1}{y}$ $\qquad = \frac{f_X(g^{-1}(y))}{\left|\frac{d}{dy}(g^{-1}(y))\right|} = \frac{f_X(g^{-1}(y))}{\left|-\frac{1}{y^2}\right|} = \frac{f_X(g^{-1}(y))}{1/y^2}$

$g(x) = \frac{1}{x}$

$g'(x) = -\frac{1}{x^2}$

$= f_X(g^{-1}(y))$

Since $x = \frac{1}{y}$

$g'(g^{-1}(y)) = -\frac{1}{(1/y)^2}$ $\qquad = f_X(1/y)$

$= y^2$ or $|g'(g^{-1}(y))| = y^2$

formula $f_Y(y) = \frac{f_X(g^{-1}(y))}{|g'(g^{-1}(y))|} = \frac{f_X(1/y)}{y^2}$

for $f_X(1/y)$ to be non zero $-1 \leq 1/y \leq 1 \Rightarrow |y| \geq 1$

so $f_X(1/y) = \frac{1}{2}$

$\therefore \; f_Y(y) = \dfrac{1}{2y^2} \quad |y| \geq 1$

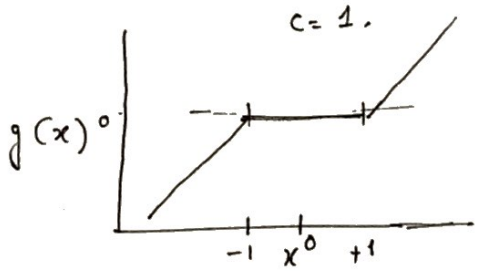
Ans

and $f_Y(y) = 0 \quad , |y| < 1$

7. X is gaussian with $\mu = 0$ & $\sigma^2 = c^2$ for some const $c > 0$

& g is
$$g(x) = \begin{cases} x - c & \text{if } x \geq c \\ 0 & \text{if } -c < x < c \\ x + c & \text{if } x \leq c. \end{cases}$$

to find $f_Y(y)$ and $F_Y(y)$ if $Y = g(X)$



when $x \geq c$ $\quad y = x - c \geq 0$ so $x = y + c$ $\quad Y > 0$

$-c < x < c$ $\quad y = 0$ so pmf at $Y = 0$

$x \leq -c$ $\quad y = x + c \leq 0$ so $x = y - c$ $\quad Y < 0$

$\therefore$ Y supports $(-\infty, 0] \cup \{0\} \cup [0, \infty)$

pdf $\quad f_X(x) = \dfrac{1}{\sqrt{c^2 2\pi}} \, e^{-x^2/2c^2}$ —— ①

~~continuous part of $f_Y(y)$ =~~

Also for CDF when $y < 0$ then $x \leq -c$

so $y = x + c \Rightarrow x = y - c$

$$\therefore \; F_Y(y) = P(Y \leq y) = P(X \leq y - c)$$
$$= \phi\left(\frac{y - c}{c}\right)$$

where ϕ is standard normal cdf.

when $y = 0$

$$F_Y(0) = P(Y \leq 0) = P(X \leq c)$$

$$= P(X \leq -c) + P(-c < X < c)$$

$$= \Phi(-1) + (\Phi(1) - \phi(-1)) = \Phi(1)$$

$y > 0 \qquad y = x - c \Rightarrow x = y + c$

$$F_Y(y) = P(Y \leq y) = P(X \leq -c \text{ or } -c \leq X < c \text{ or } c \leq X \leq y + c)$$

$$F_Y(y) = P(X \leq c) + P(c \leq X < y + c) = \Phi(1) + \left(\phi\left(\frac{y+c}{c}\right) - \phi(1)\right)$$

$$= \Phi\left(\frac{y+c}{c}\right)$$

$$\therefore F_Y(y) = \begin{cases} \phi\left(\frac{y-c}{c}\right), & y < 0 \\ \phi(1), & y = 0 \\ \phi\left(\frac{y+c}{c}\right), & y > 0 \end{cases}$$

from ①

$y < 0 \qquad f_Y(y) = \frac{d}{dy}\phi\left(\frac{y-c}{c}\right) = \frac{1}{c}\phi\left(\frac{y-c}{c}\right)$

$y > 0 \qquad f_Y(y) = \frac{d}{dy}\phi\left(\frac{y+c}{c}\right) = \frac{1}{c}\phi\left(\frac{y+c}{c}\right)$

$y = 0 \qquad P(Y = 0) = P(-c \leq X < c) = \phi(1) - \phi(-1)$

$$= 2\phi(1) - 1$$

final pdf $\quad f_Y(y) = \begin{cases} \frac{1}{c}\phi\left(\frac{y-c}{c}\right), & y < 0 \\ 2\phi(1) - 1 & y = 0 \\ \frac{1}{c}\phi\left(\frac{y+c}{c}\right) & y > 0 \end{cases}$